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Theory of polyelectrolyte complexation—Complex coacervates are self-coacervates

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The complexation of mixtures of cationic and anionic polymers to produce complex-coacervate phases is a subject of fundamental importance to colloid and polymer science as well as to applications including drug delivery, sensing technologies, and bio-inspired adhesives. Unfortunately the theoretical underpinnings of complex coacervation are widely misunderstood and conceptual mistakes have propagated in the literature. Here, a simple symmetric polyelectrolyte mixture model in the absence of salt is used to discuss the salient features of the phase diagram, including the location of the critical point, binodals, and spinodals. It is argued that charge compensation by dimerization in the dilute region renders the phase diagram of an oppositely charged polyelectrolyte mixture qualitatively and quantitatively similar to that of a single-component symmetric diblock polyampholyte solution, a system capable of "self-coacervation." The theoretical predictions are verified using fully fluctuating field-theoretic simulations for corresponding polyelectrolyte and diblock polyampholyte models. These represent the first comprehensive, approximation-free phase diagrams for coacervate and self-coacervate systems to appear in the literature. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4985568]

I. INTRODUCTION

Polyelectrolyte complexes[1–4] (PECs) are ubiquitous in natural polymers[1,5] and have long been deployed in synthetic material systems, such as coatings or encapsulants for food[5–7] and pharmaceutical[8–15] products and bio-inspired underwater adhesives.[16–20] PECs range from nanometer to micron scale discrete fluid or solid aggregates to films (e.g., polyelectrolyte multilayers[21–26]) or macroscopic phases that can be homogeneous or mesostructured. The emphasis of this article is on a specific type of PEC that can be produced by mixing together two oppositely charged polyelectrolytes in water or other polar solvents. The complex in this case is a *fluid phase*, enriched in both polymer species, that is referred to as a *complex coacervate* (CC), usually coexisting at equilibrium with a dilute "supernatant" phase.[27]

PECs can also be produced in aqueous solutions containing a *single* type of polyelectrolyte, e.g., a polycation, by introducing a second component bearing a sufficient amount of the opposite charge such as a multivalent[28] anionic counterion, nanoparticle, protein,[29–32] or colloid.[33] A particularly interesting case is a solution of a polyampholyte—a polyelectrolyte bearing charges of both signs. If the polyampholyte is "blocky," having one or more repeated units of positive charge along the backbone and one or more similar blocks of negative charge, PECs can be formed in a one-component polymer solution. Specifically, aqueous solutions of blocky polyampholytes can show *self-coacervation* phenomena: phase separation into a dense liquid CC phase rich in the polyampholyte coexisting with a dilute supernatant phase nearly devoid of polymer and containing only small counterions or salt. Self-coacervation (SC) is often discussed as a phenomenon distinct from conventional complex coacervation of two-component polyelectrolyte mixtures, but we shall see that the structure and thermodynamics of CC and SC phases are strikingly similar.

The literature on PECs is daunting because of its size, breadth of discipline (spanning physical chemistry, biochemistry, colloid science, chemical, biological, biomedical, and materials engineering, among others), and engagement of both natural and synthetic polymers in a myriad of combinations with other charged and uncharged components. In many instances, the PECs solidify upon formation and embody a non-equilibrium amorphous or crystalline state. These are the most difficult types of PECs to characterize and understand because they are at best metastable states and their structure and properties are highly path and process dependent. Here we restrict our discussion to the simpler fluid CC and SC phases under equilibrium conditions.

The earliest theories of complex coacervation, notably those by Voorn and Overbeek,[34,35] attempted to augment the veritable Flory-Huggins description of polymer solution thermodynamics with a term to account for the screened electrostatic interactions among polyions. The Voorn-Overbeek (VO) expression for the electrostatic contribution to the free energy density, F_{el}, was inspired by the Debye-Hückel theory for dilute simple (small ion) electrolyte solutions but, as will be discussed below, fails to correctly describe F_{el} for complex coacervates in any concentration regime. In spite of its conceptual and quantitative inadequacies, the VO theory continues to

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be widely applied in both experimental and theoretical studies, including very recent publications.[36–39]

The ingredients for a proper theory of complex coacervation can be found in the modern polymer physics literature, although they have not been combined previously to develop the comprehensive picture discussed here including the critical region and dilute and concentrated branches of coexistence for CC and SC formation. Specifically, Borue and Erukhimovich[40,41] pioneered the application of the random phase approximation (RPA) for concentrated systems of weakly charged polyelectrolytes, and this was further elaborated for the specific case of complex coacervation by Castelnovo and Joanny[42] and Kudlay and Olvera de la Cruz,[43] among others. The possibility of finite-sized aggregates in the dilute regime, either by ion-pairing of individual charges or by non-specific electrostatic correlations, has been considered by numerous authors including Kudlay, Ermoshkin, and Olvera de la Cruz,[44] Shusharina *et al.*,[45] Castelnovo and Joanny,[46] and Zhang and Shklovskii.[47] A recent review by Sing[48] discusses the relative merits of many of these theoretical approaches and various types of computer simulation. More recently, Qin and de Pablo[49] used a Gaussian-fluctuation analysis to highlight the importance of polymer chain connectivity to the critical point, and Shen and Wang[50] developed a renormalized Gaussian fluctuation field theory that self-consistently accounts for polymer chain conformation and connectivity in formulating a description of electrostatic correlations.

In this article, we shall focus primarily on an idealized "symmetric" polyelectrolyte mixture model capable of CC formation that is depicted in Fig. 1(a). The two linear polymers, cationic and anionic, are assumed to be flexible macromolecules of exactly the same lengths with degrees of polymerization $N_+ = N_- = N$ and equal charge densities (in units of e), $\sigma_+ = -\sigma_- = \sigma$, mixed in equal numbers $n_+ = n_- = n$ for overall electro-neutrality of the solution. The composition of their backbones is also assumed to be identical and in the absence of charge would be under near-θ conditions in water (not overly hydrophobic or hydrophilic). No counter-ions or salt is explicitly considered; a simple realization would be a mixture of a polyacid with a polybase in pure water. The solvent is treated implicitly with a uniform dielectric constant ϵ comparable to that of water. An interesting and conceptually important counterpoint to the symmetric CC model is an SC model consisting of a solution of symmetric diblock polyampholytes depicted in Fig. 1(b). Each polymer (carrying no net charge) comprises

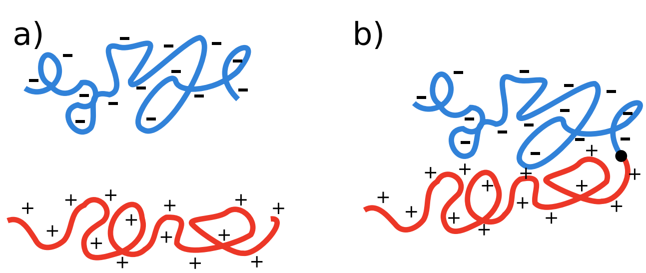

FIG. 1. (a) A symmetric homopolyelectrolyte blend composed of equal numbers of polyanions and polycations with equal degree of polymerization, N, and charge density, σ. (b) A diblock polyampholyte of length $2N$ formed by covalently linking one polyanion to one polycation.

a cationic block of length N covalently attached to an anionic block of equal length for a total polymerization degree of $2N$. The charge densities $\sigma_+ = -\sigma_- = \sigma$ are again equal and opposite. Evidently, the SC model is obtained from the CC model by linking pairs of cationic and anionic polymers in the latter to produce diblock polyampholyte molecules. While more comprehensive models with asymmetry in molecular weight, counterions, and salt can be easily constructed, the present two models shall serve to highlight the salient physics of complex and self-coacervation.

The article is organized as follows: Section II provides a conceptual theoretical analysis of the thermodynamic behavior of both CC and SC, employing asymptotic scaling arguments valid for long polymers (large N) and various regimes of concentration and charge density. Section III introduces a statistical field theory representation of the CC and SC models and presents the results of field-theoretic computer simulations. This includes both comprehensive phase diagrams and numerically derived structure factors to provide insights into the state of aggregation. Section IV provides a discussion of these results and a future outlook. A Gaussian-level analysis of the field-theory models can be found in Appendix A and numerical details of the field-theoretic simulations (FTSs) in Appendix B.

II. SCALING ANALYSIS

A. Naïve approach to a theory of complex coacervation

The simplest conceptual approach to the thermodynamic behavior of the symmetric CC model is to decompose F, the free-energy density of the solution, into separate contributions from polymer translational entropy, F_{id}, short-ranged non-electrostatic interactions, F_{sr}, and long-ranged electrostatics, F_{el}, the latter also having embedded contributions from polymer conformational entropy. For F_{id}, the expression is simply $\beta F_{id} = 2(\rho/N)\ln\rho$, where $\rho = nN/V$ is the monomer concentration of either type of polyion, 2ρ is the total monomer concentration, and $\beta = 1/(k_B T)$. In the case of F_{sr}, it is conventional to adopt a virial expansion, $\beta F_{sr} = (v/2)\rho^2 + w\rho^3$, where v and w are virial coefficients parameterizing the local two- and three-body potentials of mean force among backbone segments in implicit water. It will be assumed that v is slightly positive and $w > 0$, reflecting a solvent of slightly better than θ conditions for the polymer backbones.

It remains to discuss the form of the electrostatic free-energy density F_{el}. In the extreme limit of *infinite dilution*, $\rho \to 0$, we have an exact limiting law in the Debye-Hückel theory. The macroions in this regime are isolated, the polycations each carrying total charge $Q = \sigma N$ and the polyanions carrying charge $-Q$. The ionic strength I is proportional to $I \sim l_B(2n/V)Q^2$, where $2n/V$ is the total macroion density and $l_B = \beta e^2/\epsilon$ is the Bjerrum length consistent with Gaussian units for a uniform background dielectric of a relative dielectric constant ϵ. Re-expressed in terms of ρ and σ variables, this is $I \sim l_B\rho\sigma^2 N$, implying an *infinite dilution* Debye length

$$\xi_0 \sim \frac{1}{(l_B\rho\sigma^2 N)^{1/2}}. \qquad (1)$$

From the Debye-Hückel theory, reproduced in Appendix A, it follows that the form of F_{el} is given at infinite dilution by $\beta F_{el} \sim -1/\xi_0^3 \sim -(l_B\rho\sigma^2 N)^{3/2}$. These expressions become invalid if the macroions cannot be considered to be isolated, which is certainly the case if the unperturbed coil size approaches the Debye length. Hence, $l_B\sigma^2\rho \lesssim N^{-2}$ provides an estimated scaling bound on polyelectrolyte concentration and charge density.

In the opposite limit of a semi-dilute to concentrated mixture of weakly charged polyelectrolytes, e.g., in the complex coacervate phase where chains are well overlapping, the RPA (or "Gaussian" approximation) to an electrostatic field theory model[40,41] provides an accurate framework for computing F_{el}. This has been confirmed by field-theoretic simulations of the exact electrostatic model.[51] Specifically, in this regime, one finds that the electrostatic potential is screened not on the scale of the dilute Debye length ξ_0, but at a shorter length ξ that coincides with the geometric mean of ξ_0 and the unperturbed root-mean-square end-to-end-vector of a chain without charges, $R_0 \sim bN^{1/2}$,

$$\xi \sim (\xi_0 R_0)^{1/2} \sim \frac{1}{(l_B\rho\sigma^2 b^{-2})^{1/4}}, \tag{2}$$

where b is the statistical segment length. Crucially, the screening length ξ is independent of the macroion chain length N, as it must be because chain ends should play no role in the electrostatic screening of a fully overlapped polymer solution phase. As one would expect, the same RPA analysis (reproduced in Appendix A) suggests that in the overlapping regime the electrostatic contribution to the free energy scales as $\beta F_{el} \sim (l_B\rho\sigma^2 b^{-2})^{3/4}$.

Further insight into the physical meaning of the screening length ξ is revealed by an "electrostatic blob" argument similar to that introduced by de Gennes for one-component polyelectrolytes and by Shusharina and co-workers[45] in the context of diblock polyampholyte solutions. As depicted schematically in Fig. 2, we imagine that each chain in the solution can be viewed as a pearl necklace of blobs, each blob consisting of g sequential monomers along the chain and carrying a charge $\pm\sigma g$.

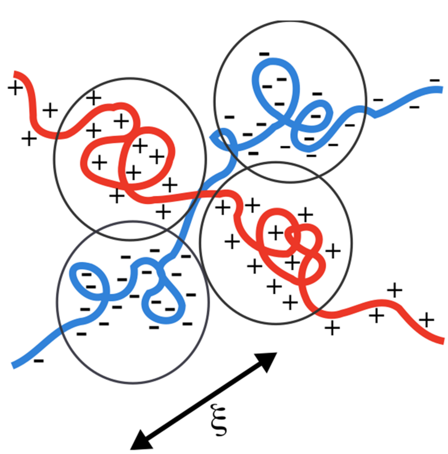

FIG. 2. Schematic of charge arrangement in symmetric polyelectrolyte blends above the overlap concentration. Blobs have electrostatic energies $\sim k_B T$ and emerge with a screening length ξ. Blobs are arranged such that positive blobs are surrounded by negative blobs to reduce electrostatic free energy.

The blobs are arranged such that positively charged blobs are largely surrounded by negatively charged blobs. The blob size ξ is set by the condition that the Coulomb energy between adjacent blobs is of the order of $k_B T$, i.e., $l_B(\sigma g)^2/\xi \sim 1$. Eliminating one of the g factors in this expression in favor of the monomer concentration $\rho \sim g/\xi^3$ and replacing the other g factor by $(\xi/b)^2$, which follows from random-walk statistics within a blob (recall the near-θ solvent conditions), we find that the scaling expression for ξ given in Eq. (2) is exactly reproduced. Comparing Eqs. (1) and (2), it is evident that the crossover from dilute to semi-dilute/concentrated conditions occurs at $\rho \sim 1/(l_B\sigma^2 N^2 b^2)$, which occurs when $\xi \sim \xi_0 \sim R_0$ as might be expected. Under extremely dilute conditions, the appropriate screening length ξ_0 is much larger than R_0, while in the overlapping regime the screening length ξ, which is N-independent, is much smaller than $R_0 \sim bN^{1/2}$ for large polymers ($N \gg 1$).

With this basic picture in hand and expressions for F_{el} appropriate in the various concentration regimes, it is useful to reflect on the classic Voorn-Overbeek (VO) model for the electrostatic energy density. The VO model is based on the expression $\beta F_{el} \sim -(l_B\rho\sigma)^{3/2}$, which we see is *incorrect* both in the extreme dilute limit (with the wrong power of σ and missing N dependence) and in the concentrated regime (with incorrect exponents for all variables except N). The VO model is based on a picture where all the bound polymer ions, of concentration $\rho\sigma$, are detached from the polymers and their electrostatic correlation energy is computed using the Debye-Hückel theory. Obviously the fact that the charges are bound to the polyelectrolyte backbones is an essential piece of physics that cannot be ignored. It is surprising that this major deficiency in the VO theory has not been appreciated by the colloid and polymer science communities and continues to be applied in the modern literature. A recent contribution by Perry and Sing[52] argues that although the VO theory makes strong errors in structure and charge aggregation, this is somewhat compensated by excluded-volume effects in some limits. Our opinion, however, is that the VO theory is entirely unsatisfactory, both from a conceptual standpoint and from a quantitative standpoint.

Based on the scaling picture above, we can construct a tentative phase diagram by examining the chemical potential, $\mu = dF/d\rho$, and osmotic pressure, $\Pi = \rho\mu - F$ of the polyelectrolyte mixture. Specifically, the osmotic pressure is

$$\beta\Pi = \frac{2\rho}{N} + \frac{1}{2}v\rho^2 + \beta\Pi_{el}(\rho), \tag{3}$$

where $\beta\Pi_{el}$ has the same limiting scaling dependences on ρ, N, and σ as βF_{el}.[85] The critical point, located by $\partial\Pi/\partial\rho = \partial^2\Pi/\partial^2\rho = 0$, is easily confirmed to be located in the dilute regime where $\rho l_B\sigma^2 b^2 \ll N^{-2}$, specifically with $\rho_c \sim 1/(vN)$ and $\sigma_c \sim (v^{1/3}/l_B)^{1/2} N^{-2/3}$. A phase diagram in the coordinates of σ versus ρ would have a two-phase region of liquid-liquid coexistence expanding out of the critical point for $\sigma > \sigma_c$ and a homogeneous single phase region for $\sigma < \sigma_c$. The dilute (supernatant) and concentrated (complex coacervate) branches of the coexistence curve (binodal) can be obtained by equating μ and Π in the two phases.

The dilute phase proves to be extraordinarily dilute at large N and is dictated in scaling by $\rho^I \sim \exp[-Nv^{1/5}(l_B\sigma^2/b^2)^{3/5}]$. In turn, the concentrated branch, ρ^{II}, can be estimated by this diluteness assumption, $\Pi^{II} = \Pi^I \approx 0$, which leads to $\rho^{II} \sim v^{-4/5}(l_B\sigma^2/b^2)^{3/5}$.

B. Dimers and self-coacervation

The above picture for the behavior of the CC model seems reasonable, until we consider the possibility that in the dilute phase, individual polycations and polyanions could *pair to form charge neutral globules*, lowering the electrostatic correlation energy in return for less translational entropy. As argued by de Gennes,[53] an isolated polyanion or polycation would adopt a linear (rod-like) configuration containing $N_b = N/g$ electrostatic blobs each of size $\xi \sim (b^4/l_B)^{1/3}\sigma^{-2/3}$. The electrostatic correlation energy gained by combining, or dimerizing, a polyanion/polycation pair into a charge neutral globule, depicted in Fig. 3, should be of order $-k_BT2N_b$, since the $2N_b$ blobs can arrange themselves to ensure that adjacent electrostatic blobs are of opposite sign. This suggests that the free energy change for dimerization ΔF_d should scale as

$$\beta\Delta F_d \sim -2N_b \sim -N/g \sim -(l_B/b)^{2/3}\sigma^{4/3}N. \tag{4}$$

Treating the dimerization process as a chemical equilibrium, we expect that the density of dimers C_d should be related to the density of unbound polyelectrolytes, $C_u = \rho/N$, by the formula

$$C_d = C_u^2 V_c \exp(-\beta\Delta F_d), \tag{5}$$

where V_c is a capture volume estimated as $V_c \sim N_b\xi^3 \sim Nb^3(b/(l_B\sigma^2))^{1/3}$.

From Eqs. (4) and (5), we see that pairing into dimers becomes significant when $\sigma > \sigma_d$, with $\sigma_d \sim N^{-3/4}$. Because $\sigma_d < \sigma_c$ for large N, it is evident that we have a *contradiction*—dimers should already be present in the critical region and increasingly so at larger σ values, yet we have estimated the location of the critical point by ignoring them. A simple way to rectify the situation is to explicitly pair all polycations with a polyanion creating a solution of diblock polyampholytes. Apart from the covalent bond that suppresses the translational entropy of the paired polyanion and polycation ends [a weak $O(1/N)$ effect], an isolated globule of such a diblock polyampholyte should be structurally and thermodynamically very similar to a dimer globule in the polyelectrolyte mixture. We thus proceed with a theory for a solution of diblock polyampholytes; this is nothing more than the SC model described in the Introduction.

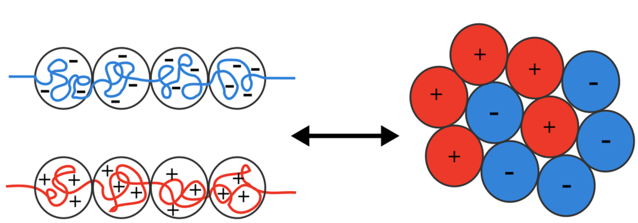

FIG. 3. Two-dimensional schematic of the rod-to-globule microstructure transition in dilute polyelectrolyte solutions.

As demonstrated in Appendix A, the electrostatic contribution to the free energy of a diblock polyampholyte solution is no longer nonanalytic in monomer concentration ρ because the polyampholytes carry no net charge. In the extreme dilute regime, two isolated globules interact electrostatically by a fluctuating dipole/induced dipole London-type mechanism with an r^{-6} distance dependence. The overall second virial coefficient is thus finite, unlike the case of a simple salt solution or a solution of unpaired polyanions and polycations. It is shown in Appendix A that the leading low concentration contribution to the electrostatic excess free energy of a polyampholyte solution is indeed of the second virial (quadratic in ρ) form

$$\beta F_{el}(\rho) \sim -b(l_B\sigma^2)^2N^{5/2}\rho^2. \tag{6}$$

Including the ideal translational entropy and short-range nonelectrostatic interactions, we have the following expression for the free energy density of a polyampholyte solution:

$$\beta F = \frac{\rho}{2N}\ln\rho + \frac{1}{2}[v - b(l_B\sigma^2)^2N^{5/2}]\rho^2 + w\rho^3 + \cdots, \tag{7}$$

which identifies $v_{\rm eff} = v - b(l_B\sigma^2)^2N^{5/2}$ as an effective second-virial coefficient. The thermodynamic analysis of this expression now follows standard lines;[54] the critical value of σ corresponds to the vanishing of $v_{\rm eff}$ (the so-called θ condition), which implies $\sigma_c \sim (v/(bl_B^2))^{1/4}N^{-5/8}$. Thus, the critical charge density has been shifted up considerably from the naïve theory where $\sigma_c \sim N^{-2/3}$. The critical concentration has also shifted to larger ρ, scaling as $\rho_c \sim \rho^\star \sim N^{-1/2}$, (where $\rho^\star$ is the overlap concentration) similar to the case of solutions of an uncharged neutral polymer near the θ point.

Figure 4 summarizes these results by contrasting in schematic form the predictions of the naïve theory and the dimer polyampholyte theory. In summary, we have argued that a symmetric mixture of polycations and polyanions, i.e., the CC model, should have a phase diagram and thermodynamic properties qualitatively and quantitatively similar to that of a single component solution of symmetric diblock polyampholytes at double the molecular weight, namely, the SC model.

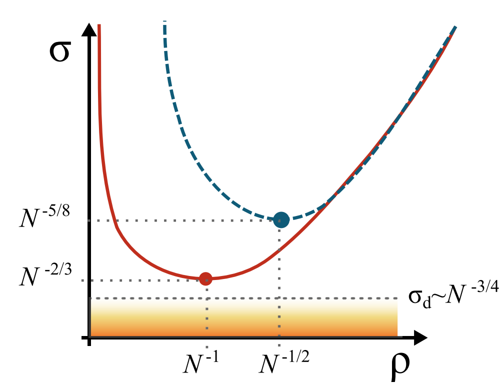

FIG. 4. Schematic phase coexistence curves generated from scaling arguments. Red solid line—naïve result for a symmetric polyelectrolyte blend, with critical concentration (red circle) $\rho_c \sim N^{-1}$ and critical charge density $\sigma_c \sim N^{-2/3}$. Blue dashed line—polyampholyte with critical composition (blue circle) $\rho_c \sim N^{-1/2}$ and critical charge density $\sigma_c \sim N^{-5/8}$. The orange shaded area denotes a region in which non-dimerized polyelectrolytes are abundant.

III. FIELD-THEORETIC SIMULATIONS

To validate the schematic picture just outlined, we have conducted field-theoretic simulations (FTSs) of polyelectrolyte coacervation for models[51,55–57] analogous to those discussed in Sec. II. Determining the conditions of phase coexistence requires access to thermodynamic properties across many orders of magnitude of polyelectrolyte concentration. Field-theoretic simulations using complex Langevin (CL) sampling provide an approximation-free approach for computing such properties, which is especially efficient for dense systems of overlapping high-molecular-weight polymers. As we show in this section, extremely dilute conditions can also be accessed within the same FTS framework, thus allowing for simulations covering the large span of concentrations required for constructing phase diagrams.

The coarse-grained model of interest is an assembly of continuous Gaussian chains with statistical segments smeared over a finite volume by convolving with a normalized Gaussian profile. The smearing process ensures that pressures and chemical potentials computed in our simulations are insensitive to the computational grid and free of ultra-violet divergences.[58–60] As in the scaling analysis in Sec. II, all pairs of statistical segments interact through a weak contact excluded volume parameter, v, while charged segments interact via a Coulomb potential screened by a uniform background dielectric with Bjerrum length l_B. The resulting interaction energy is $\beta U_{int} = \frac{v}{2}\int d\mathbf{r}\,\bar{\rho}^2(\mathbf{r}) + \frac{l_B}{2}\int d\mathbf{r}\int d\mathbf{r}'\,\frac{\bar{\rho}_e(\mathbf{r})\bar{\rho}_e(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} - \beta U_{SI}$. The microscopic density of segment centers is $\hat{\rho}(\mathbf{r}) = \sum_i^n \int ds\,\delta(\mathbf{r}-\mathbf{r}_i(s))$ for n polymer chains (where s is a continuous backbone contour variable and $\mathbf{r}_i(s)$ is a space curve encoding the configuration of polymer chain i) and its smeared analog is $\bar{\rho}(\mathbf{r}) = \int d\mathbf{r}'\,\Gamma(|\mathbf{r}-\mathbf{r}'|)\,\hat{\rho}(\mathbf{r}')$, where $\Gamma(r) = \left(2\pi a^2\right)^{-3/2}\exp\left(-r^2/2a^2\right)$ is a normalized Gaussian smearing function of width a. Similarly, the electrostatic charge density is $\hat{\rho}_e(\mathbf{r}) = \sum_i^n \int ds\,\sigma_i(s)\,\delta(\mathbf{r}-\mathbf{r}_i(s))$, with smeared analog $\bar{\rho}_e(\mathbf{r})$, where $\sigma_i(s)$ is the segmental charge valence at position s on polymer chain i. Finally, βU_{SI} is a self-interaction correction for overcounting self-segment interactions in the other two terms.

Following the methods outlined in Ref. 55, the canonical partition function of this model, integrated over segment coordinates, can be converted via an exact transformation to a complex-valued statistical field theory,

$$\mathcal{Z}_c = \frac{Z_0}{Z_w Z_\varphi}\int \mathcal{D}w \int \mathcal{D}\varphi\, e^{-H[\varphi,w]}, \tag{8}$$

where Z_0 contains the partition function for an ideal gas of continuous Gaussian chains and self-interaction corrections, and the field-theoretic Hamiltonian is

$$H[\varphi,w] = \frac{1}{2B}\int d\mathbf{r}\,w^2(\mathbf{r}) + \frac{1}{2E}\int d\mathbf{r}\,|\nabla\varphi(\mathbf{r})|^2 - \sum_l \frac{CV\bar{\phi}_l}{R_g^3\alpha_l}\ln Q_l[\varphi,w;\bar{a}]. \tag{9}$$

The field φ can be interpreted as a fluctuating electrostatic potential necessary to decouple the Coulomb interactions, while w is a fluctuating auxiliary potential that decouples the excluded volume interactions. The long-range nature of the Coulomb interaction is preserved by fluctuations in the φ field, even though explicit non-locality in the field theory is restricted to a semi-local gradient term in the Hamiltonian.

In our field-theory simulations, it proves convenient to rescale model parameters to a minimal set of dimensionless quantities by reducing chain polymerization degrees by a reference N to $\alpha_l = N_l/N$, where l indexes polymer chain *types*, and scaling all spatial lengths by a corresponding ideal homopolymer radius of gyration $R_g = \left(b^2N/6\right)^{1/2}$. Polymer molecular weight can be varied independently of other parameters in the model by changing α_l with N fixed. This scaling step yields the dimensionless parameters present in the Hamiltonian of Eq. (9): $\bar{a} = a/R_g$, a smearing scale for polymer segments; $B = vN^2R_g^{-3}$, a measure of excluded volume/implicit solvent quality; $C = \sum_l n_l\alpha_l R_g^3/V = \rho R_g^3/N$, a polymer chain number density; and $E = 4\pi l_B\sigma^2N^2Rg^{-1}$, a measure of electrostatic interaction strength, where σ is a reference chain charge density. These four parameters, together with polymer chain lengths α_l, the overall fraction of segments derived from each chain type $\bar{\phi}_l = n_l\alpha_l(\sum_m n_m\alpha_m)^{-1}$, and segmental charge valences along the backbone of each polymer chain type l, are sufficient to fully specify the system.

The two systems of initial interest for this study are a symmetric polyelectrolyte (PE) blend (two chain types: $\alpha_+ = \alpha_- = 1$ and $\bar{\phi}_+ = \bar{\phi}_- = 1/2$) and a symmetric diblock polyampholyte (PA) formed by joining each polyanion to a polycation (one chain type: $\alpha = 2$ and $\bar{\phi} = 1$). Note that since the two models share the same reference N, equal C parameters between the two models imply equal monomer concentrations, while equal E implies equal charge density and background dielectric screening. We initially fix $B = 1$ and $a = 0.2\,R_g$ and defer a discussion on the role of these model parameters until later.

As a consequence of the transformation to a field theory, statistical segments are decoupled and interact only with auxiliary Hubbard-Stratonovich fields $w(\mathbf{r})$ and $\varphi(\mathbf{r})$. The statistics of polymer chain type l subject to these fields is described by the single-molecule partition function $Q_l[\varphi,w] = V^{-1}\int d\mathbf{r}\,q_l(\mathbf{r},\alpha_i;[\varphi,w])$, where q_l is a chain propagator that, for continuous Gaussian chains, satisfies a modified diffusion equation of the form

$$\frac{\partial}{\partial s}q_l(\mathbf{r},s) = \left[\nabla^2 - \psi_l(\mathbf{r},s)\right]q_l(\mathbf{r},s) \tag{10}$$

with initial condition $q_l(\mathbf{r},s=0) = 1$. The contour-position-dependent species chemical potential field $\psi_l(\mathbf{r},s)$ entering Eq. (10) is given by $\psi_l(\mathbf{r},s) = i\Gamma \star \left(w + \frac{\sigma_l(s)}{\sigma}\varphi\right)$, where $i = \sqrt{-1}$, $\star$ denotes a spatial convolution, and in the present model specializations with fixed magnitude of charge density, $\sigma_l(s)/\sigma$ reduces to $+1$ for cationic segments and -1 for anionic segments. A similar propagator defined for traversing the polymer contour variable in reverse is useful for constructing segment density operators and field derivatives of H,

$$\frac{\partial}{\partial s}q_l^\dagger(\mathbf{r},s) = \left[\nabla^2 - \psi_l(\mathbf{r},\alpha_l - s)\right]q_l^\dagger(\mathbf{r},s) \tag{11}$$

with initial condition $q_l^\dagger(\mathbf{r},s=0) = 1$.

In addition to the field-theoretic partition function, operators must be developed that, when averaged over the ensemble of field configurations, yield the corresponding thermodynamic observables. In the present work, we are largely concerned with chemical potential, $\beta\mu_l = \langle\tilde{\mu}_l\rangle$, and pressure, $\beta\Pi R_g^3 = \left\langle\tilde{\Pi}\right\rangle$, where

$$\langle\ldots\rangle = \frac{\int \mathcal{D}w \int \mathcal{D}\varphi \ldots \exp(-H[\varphi, w])}{\int \mathcal{D}w \int \mathcal{D}\varphi \exp(-H[\varphi, w])} \tag{12}$$

denotes a field-based thermal average. As discussed previously,[60,61] the normalizing denominators Z_w and Z_φ that are introduced by Hubbard-Stratonovich transformations[55] must be consistently incorporated into the evaluation of thermodynamic observables. For the present work, these objects, which depend on the system volume but not on its composition, affect only on how the pressure is evaluated. Additionally, ideal-gas contributions from Z_0 must be properly included in all operators for the correct establishment of chemical and mechanical equilibrium between coexisting phases of differing concentrations. The chemical potential operator for chain type l, including the ideal-gas translational entropy contribution, is $\tilde{\mu}_l = \ln\frac{C\bar{\phi}_l}{\alpha_l} - \ln Q_l[\varphi, w]$. Equation (B1) of Appendix B shows the efficient low-variance[61] $\tilde{\Pi}$ operator used in this work.

The field-theory model discussed here is complex-valued.[55] Although the functional integrals in Eq. (8) are defined to be over real-valued $w(\mathbf{r})$ and $\varphi(\mathbf{r})$ fields, those fields enter the modified diffusion equation for single-chain statistics, Eq. (10), with imaginary coefficients. As a result, the exponentiated H functional in Eq. (8) possesses a rapidly oscillating phase that leads to a critical efficiency loss for traditional sampling methods known as the sign problem. By promoting the fields to be complex variables and employing complex Langevin (CL) sampling,[62,63] the efficiency loss is avoided. The CL equations of motion used here are

$$\frac{\partial w(\mathbf{r}, t)}{\partial t} = -\lambda_w \frac{\delta H[\varphi, w]}{\delta w(\mathbf{r}, t)} + \eta_w(\mathbf{r}, t), \tag{13}$$

$$\frac{\partial \varphi(\mathbf{r}, t)}{\partial t} = -\lambda_\varphi \frac{\delta H[\varphi, w]}{\delta \varphi(\mathbf{r}, t)} + \eta_\varphi(\mathbf{r}, t), \tag{14}$$

where the $\eta_i(\mathbf{r}, t)$ are real-valued Gaussian-distributed white noise random variables with statistics $\langle\eta_i(\mathbf{r}, t)\rangle = 0$, $\langle\eta_i(\mathbf{r}, t)\eta_i(\mathbf{r}', t')\rangle = 2\lambda_i\delta(\mathbf{r} - \mathbf{r}')\delta(t - t')$. Numerically propagating the CL equations of motion generates importance-sampled sequences of field configurations, and ensemble averages over field-theoretic operators can be replaced by time averages over field samples by the ergodic principle. In the present work, we sample the CL dynamics using the exponential time differencing algorithm.[59,61,64] Further CL numerical details are described in Appendix B.

The field-theoretic model can also be analyzed theoretically with perturbation theory about the homogeneous state. In this case, we use a standard Gaussian approximation, or "random phase approximation" (RPA), for $\mathcal{Z}_c$. Appendix A shows the resulting approximate expressions for $\beta\mu$ and $\beta\Pi R_g^3$ for the systems of interest, together with a numerical comparison between the RPA observables and those computed with CL. The excellent agreement at high concentration, where field fluctuations are weak and RPA is expected to be good, provides a validation of both the Gaussian analysis and the numerical quality of the CL simulations.

A. Determination of phase equilibria

The method used here for constructing phase equilibrium conditions involves explicit computation of $R_g^3\beta\Pi(C)$ and $\beta\mu(C)$ for a range of concentrations C. Figure 5(a) shows a parametric plot of $R_g^3\beta\Pi(C)$ vs. $\beta\mu(C)$ at fixed $E = 400$ for the symmetric diblock polyampholyte, with each point obtained for a different C value. The data fall into three groups: a dilute low-C one-phase branch with low osmotic pressure, a concentrated high-C one-phase branch, and an unstable branch. Recall that the simulations are conducted in the canonical ensemble; those simulations constrained to concentrations that fall inside the spinodal envelope will

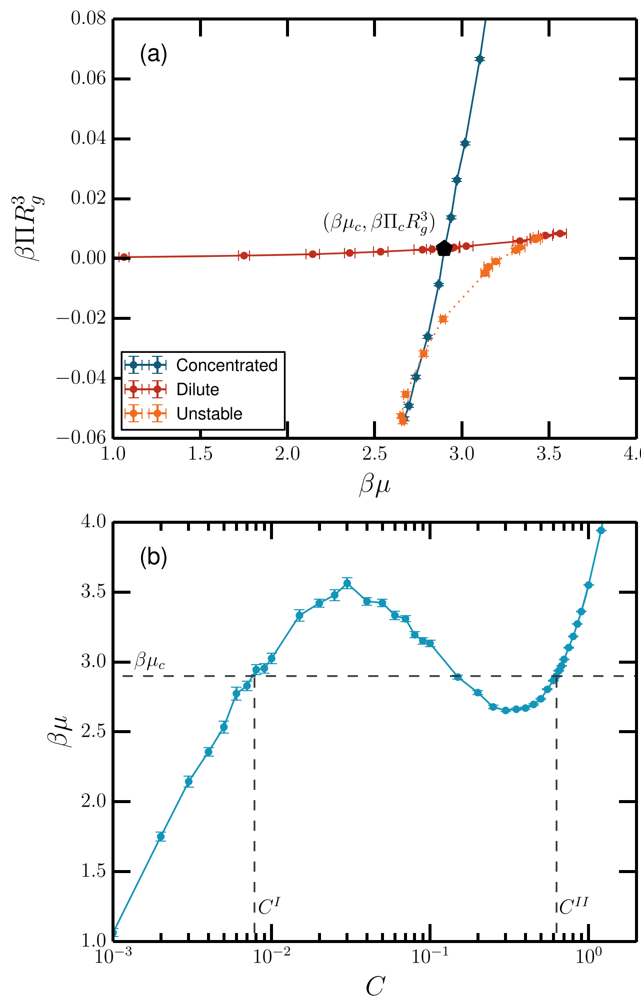

FIG. 5. Determination of phase equilibrium conditions for a symmetric diblock polyampholyte with $\bar{a} = 0.2$, $B = 1.0$, and $E = 400$. (a) Overlaid $\beta\mu$-$\beta\Pi R_g^3$ curves for concentrated (CC) and dilute (supernatant) phases show conditions of simultaneous matched chemical potential and pressure. (b) Intersection of chemical potential versus dimensionless polyampholyte concentration with $\beta\mu_c$ determines supernatant (C^I) and self-coacervate (C^{II}) coexisting phase concentrations.

experience phase separation into two domains during the simulation. In the thermodynamic limit, where interfacial contributions are irrelevant, and with ergodic sampling, one would expect from chemical and mechanical stabilities between the domains that $\beta\mu = \beta\mu_c$ and $\beta\Pi R_g^3 = \beta\Pi_c R_g^3$ for all compositions inside the binodal. The orange points of Fig. 5(a) contain relatively large finite-size errors causing a deviation from this limiting behavior. However, we stress that simulations within the spinodal envelope are not required or used to construct binodals. By extracting the coordinate of crossing between concentrated and dilute branches on this plot, we determine the osmotic pressure $\beta\Pi_c R_g^3$ and chemical potential $\beta\mu_c$ at which chemical and mechanical equilibrium conditions can be established between coexisting coacervate and supernatant phases. A seemingly more efficient way to uncover the coexistence conditions is provided by sampling in the Gibbs ensemble.[65] However, the extreme diluteness of the supernatant phase in the present case renders stochastic sampling of concentration challenging.

Once $\beta\mu_c$ and $\beta\Pi_c R_g^3$ are determined, the construction shown in Fig. 5(b) yields the binodal concentrations. Combining such extractions for various E results in the phase diagrams shown in Fig. 6. The procedure can be repeated using RPA estimates (Appendix A) of the chemical potential and pressure. From Fig. 6, we make the following observations: (1) higher-order fluctuation corrections present in approximation-free CL simulations shift the critical point to higher E compared to RPA theory. (2) Errors in low-C values of $\beta\mu$ from RPA theory result in a multiple-order-of-magnitude difference in predicted polyelectrolyte concentration in the supernatant phase. However, the compositions of the coacervate phases are in surprisingly close agreement between FTS-CL and RPA. (3) The close agreement between PE and PA phase diagrams reinforces the perspective that complex coacervates are thermodynamically analogous to block polyampholyte self-coacervates. We emphasize that the FTS-CL results shown in Fig. 6 represent the first "exact" phase diagrams, involving no uncontrolled approximations, to be generated for a model coacervate or self-coacervate system.

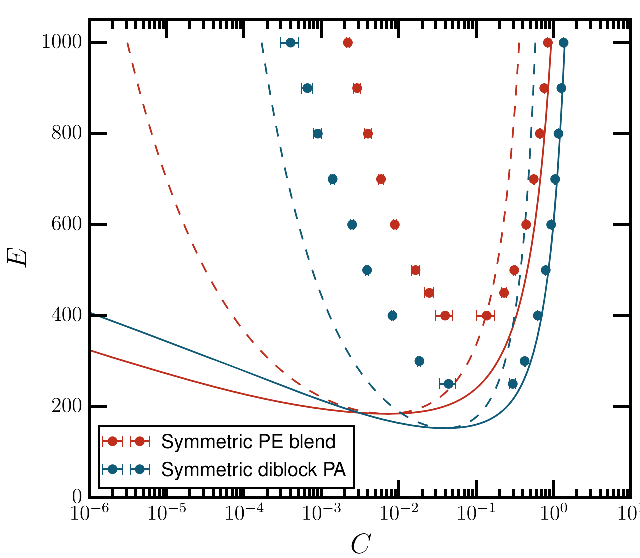

FIG. 6. Phase diagram for a symmetric polyelectrolyte (PE) blend (red) and an equivalent diblock polyampholyte (PA, blue) with the same total molecular weight and monomer concentration. In both cases, $\bar{a} = 0.2$ and $B = 1$. Phase coexistence (binodal) boundaries are shown as points for FTS and solid lines for RPA theory. Dashed lines show the RPA single-phase stability limits (spinodals).

B. Critical point scaling

According to the scaling arguments in Sec. II, one should expect the PE dilute state in FTS-CL simulations to consist almost entirely of spontaneously paired polyanions and polycations. This nature is unlikely to be reflected in the random phase approximation, which shows Debye-Hückel-like scaling at moderately low concentrations. For consistency with the scaling arguments, *both* PE and PA systems should have a critical point that varies as $\rho_c \sim N^{-1/2}$ and $\sigma_c \sim N^{-5/8}$. Converting to the dimensionless parameters used in our RPA and FTS-CL analyses, these predictions correspond to $C_c \sim \alpha^{-1/2}$ and $E_c \sim \alpha^{-5/4}$, where the reduced degree of polymerization α can be manipulated to affect the proportional change in molecular weight. Figure 7 shows the FTS-CL binodal envelopes in the E-C plane for various molecular weights for PE [panel (a)] and PA systems [panel (b)]. The scaling of E_c with respect to molecular weight is shown in Fig. 8 to follow a −5/3 power law for both PE and PA systems, providing further evidence that complex coacervate and self-coacervate systems are analogous in the critical region of the phase diagram. However, the observed scaling $E_c \sim \alpha^{-5/3}$ differs from the predictions in Sec. II. Furthermore, we note that this predicted scaling corresponds to $\sigma_c \sim N^{-5/6}$, which is a faster decrease with molecular weight than the dimerization charge density, σ_d, predicted in our scaling analysis. In the following, we discuss potential causes of these discrepancies.

Localization of the critical point from FTS-CL binodals is challenging using the techniques presented here due to increasing fluctuation strength (population variance) in $\beta\tilde{\mu}$ and $\beta\tilde{\Pi}$ operators as the critical point is approached. This difficulty is compounded by the narrowing of the domain of metastability of one-phase regions as binodal and spinodal envelopes approach each other. Furthermore, correlation lengths increase in the critical region. Regardless of an earlier literature uncertainty on whether simple ionic fluids follow Ising-like or classical critical exponents,[66–71] we expect correlation lengths to diverge at the critical point in this system, leading to larger finite-size errors in the region of the critical point. These numerical difficulties are all enhanced as the molecular weight is reduced, which may lead to bias in the extracted molecular-weight dependence of E_c.

Alongside FTS-CL numerical challenges, the scaling analysis in Sec. II contains sources of potential bias. The analysis used the assumption that electrostatic blobs were well formed throughout the relevant concentration and charge-density ranges, and that the critical region was sufficiently dilute to follow dilute-PA scaling of the electrostatic free energy. Finally, the RPA itself, which is used to extract dilute and concentrated scaling laws for the electrostatic free energy, may be unreliable in the critical region. Despite some uncertainty on the precise numerical value of the scaling exponents, we re-emphasize the crucial observation from the

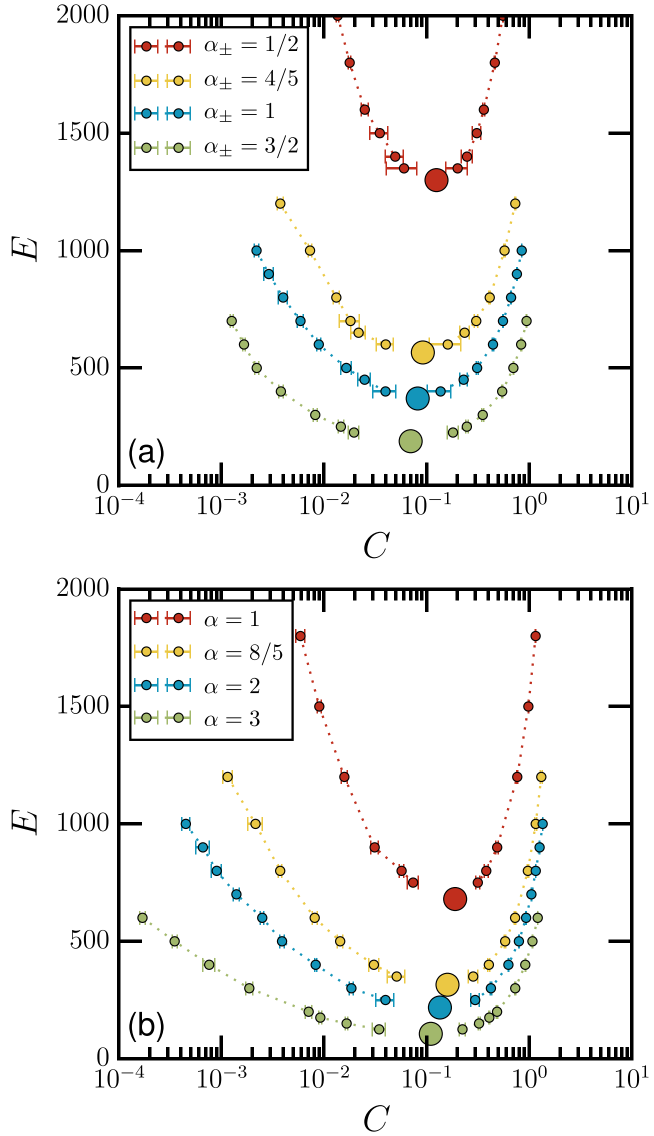

FIG. 7. FTS-CL binodal envelopes and estimated critical points (large circles) for a symmetric polyelectrolyte blend [panel (a)] and a symmetric diblock polyampholyte [panel (b)] for a variety of molecular weights.

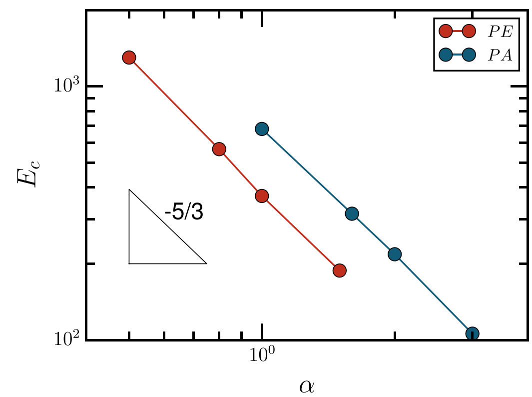

FIG. 8. Variation of critical E with molecular weight, α, indicates a scaling law $E_c \sim \alpha^{-5/3}$ for both PE and PA systems.

simulations that E_c and C_c scale analogously for PE and PA systems.

The results presented herein were generated with a fixed smearing scale $\bar{a} = 0.2$. The value of this model parameter influences the phase diagram, as demonstrated in the supplementary material by comparing FTS-CL binodal envelopes of $\bar{a} = 0.15$, 0.2, 0.25, and 0.3. It is readily demonstrated that a similar type of variation in the phase diagram can be achieved by modifying B with $\bar{a}$ fixed. Since $\bar{a}$ is responsible for locally smearing the microscopic density operators, which is equivalent to replacing a contact interaction between point particles with a soft Gaussian potential,[58,60,72] it might be anticipated that $\bar{a}$ and B are strongly coupled and the primary thermodynamic role of $\bar{a}$ is to renormalize the strength of the short-range excluded-volume repulsion between polymer segments. If this is the case, one could anticipate that holding constant a suitably renormalized excluded-volume parameter, $B_r(\bar{a}, B, C)$, would render the computed binodals independent of $\bar{a}$. A procedure for constructing B_r might involve a Hartree[73–76] or self-consistent-one-loop approximation based on the excluded-volume loop integrals of Eqs. (A2) and (A5). The topic of model universality, independence of mesoscale predictions to micro-scale model details, and interpretation of UV-regularized smeared models in terms of the parameters of non-smeared models remains an active area of research.[60,77] In the present work, we are interested in coacervation phenomena and not with making quantitative connection with experiments; thus we fixed $\bar{a}$ and B to constant values in constructing phase diagrams. However, we note that it is straightforward to adopt any parameterized form for the excluded-volume interaction, including any concentration dependence, with the caveat that predicted critical-point scalings may be affected if B depends on C.

C. Simulation evidence for dilute-phase dimerization

By constructing the RPA and CL binodals and extracting critical scaling exponents in Secs. III A and III B, we reinforced the thermodynamic analogy between PE complex coacervation and PA self-coacervation. However, the phase diagrams do not provide a direct view of polyelectrolyte complexation, and specifically the possibility of dimerization, in the dilute phase. To determine the aggregation microstructure of the dilute phase, we evaluate the particle-center-coordinate structure factor for polymer-segment opposite-charge correlations, $S_{+-}(\mathbf{k})/\rho N = \frac{V}{CN^2}\langle \delta\hat{\rho}_+(\mathbf{k})\,\delta\hat{\rho}_-(-\mathbf{k})\rangle$, where $\delta\hat{\rho}_i(\mathbf{k}) = \hat{\rho}_i(\mathbf{k}) - \langle\hat{\rho}_i(\mathbf{k})\rangle$. An efficient field-theoretic operator for this object, $\tilde{S}_{+-}/\rho N$, is provided in Eq. (B2). Evaluating the cumulant moment of the $\tilde{S}_{+-}$ operator yields the structure factor of interest: $S_{+-}(\mathbf{k})/\rho N = \left\langle \tilde{S}_{+-}(\mathbf{k})/\rho N\right\rangle_c$, where $\langle f(\mathbf{k})\,g(-\mathbf{k})\rangle_c = \langle f(\mathbf{k})\,g(-\mathbf{k})\rangle - \langle f(\mathbf{k})\rangle\langle g(-\mathbf{k})\rangle$.

The random-phase approximation (RPA) estimate of the opposite-charge, particle-center structure factor, which is expected to be accurate for high C, is

$$\frac{S_{+-}^{RPA}(\mathbf{k})}{\rho N} = \frac{1}{4\rho N}\left(S_n^{RPA}(\mathbf{k}) - S_e^{RPA}(\mathbf{k})\right), \qquad (15)$$

where the neutral ($S_n = \langle\delta\hat{\rho}(\mathbf{r})\,\delta\hat{\rho}(\mathbf{r}')\rangle$) and electrostatic ($S_e = \langle\delta\hat{\rho}_e(\mathbf{r})\,\delta\hat{\rho}_e(\mathbf{r}')\rangle$) structure factors for the symmetric polyelectrolyte blend with $\alpha_+ = \alpha_- = 1$ are given by[51]

$$\frac{S_{n,PE}^{RPA}(\mathbf{k})}{\rho N} = \frac{g_D\left(k^2\right)}{1 + BCg_D\left(k^2\right)V^2\hat{\Gamma}^2(\mathbf{k})}, \tag{16}$$

$$\frac{S_{e,PE}^{RPA}(\mathbf{k})}{\rho N} = \frac{k^2 g_D\left(k^2\right)}{k^2 + ECg_D\left(k^2\right)V^2\hat{\Gamma}^2(\mathbf{k})}, \tag{17}$$

and the corresponding expressions for the symmetric diblock polyampholyte with $\alpha = 2$ are

$$\frac{S_{n,PA}^{RPA}(\mathbf{k})}{\rho N} = \frac{h_+\left(k^2\right)}{1 + BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(\mathbf{k})}, \tag{18}$$

$$\frac{S_{e,PA}^{RPA}(\mathbf{k})}{\rho N} = \frac{k^2 h_-\left(k^2\right)}{k^2 + ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(\mathbf{k})}, \tag{19}$$

with Debye functions g_D, h_+, and h_- defined in Appendix A, and $V^2\hat{\Gamma}^2(\mathbf{k}) = e^{-\bar{a}^2k^2}$. Similar to the discussion in Ref. 51, the electrostatic structure factors $S_{e,PE}^{RPA}$ and $S_{e,PA}^{RPA}$ can be approximated at high k by $\frac{k^2\rho N}{S_e^{RPA}(\mathbf{k})EC} \approx e^{-\bar{a}^2k^2} + k^4\xi^4$, where $\xi = (2EC)^{-1/4}$ is the same electrostatic correlation ("screening") length introduced in Eq. (2). ξ and $\bar{a}$, the scale of segment smearing, are the only two physically relevant length scales in a dense homogeneous phase of overlapping polyelectrolyte chains. The fact that the RPA screening length ξ is appropriate for both the PE and PA models, arising from the equal high-k expansions of g_D and h_-, reinforces the unimportance of chain ends and polyampholyte block connectivity at short length scales in overlapping configurations.

Figure 9 shows dilute-phase $S_{+-}(\mathbf{k})/\rho N$ data for the PE symmetric polyelectrolyte blend and the equivalent PA symmetric diblock polyampholyte, both evaluated with FTS-CL and RPA. In the idealized limit $E \to 0$, charges are well screened by the implicit background dielectric and opposite-charge correlations can only exist through polymer-chain connectivity (PA model) or diffuse Debye-Hückel correlations (PE model) with a divergent screening length [cf. Eq. (1)]. The fact that the PA system contains connected opposite charges is manifest in the non-zero low-E structure factor, which is well matched between RPA and CL. As E is increased (screening reduced), the PA RPA structure factor is largely unchanged, but the CL structure factor exhibits enhanced correlations at larger k or shorter spatial distances. We attribute this short-length-scale responsiveness to strong local correlations that emerge within the isolated polyampholyte globules due to monomer-scale charge ordering driven by an associated reduction in electrostatic free energy. These correlations are not captured by the RPA theory.

The PE blend shows a different trend with variation of E. For very small $E \ll E_c$, e.g., $E = 5$, the dilute phase comprises a Debye-Hückel-like solution of isolated polyanions and polycations. Consequently, both RPA and CL data show negligible opposite-charge correlations up to near-macroscopic length scales coinciding with the Debye length, Eq. (1). As the screening length is reduced by increasing E, we observe an increase in CL-predicted opposite-charge correlations that matches a scaled version of the *polyampholyte diblock* RPA prediction. We attribute this to fractional dimerization of the disordered polyelectrolyte coils. For example, $E = 100$, still far below the critical value E_c, shows approximately 60% statistical dimerization. Finally, for high-E, the dimers present in the dilute PE phase undergo the same charge-ordered globule transition as found in the dilute PA, and the structure factors of the two systems are largely indistinguishable.

It is interesting to contrast the structure factors in dense overlapping configurations. Figure 10 shows dense-phase ($C = 10$) opposite-charge structure factors for both the PE and PA models. Here, as expected, no difference is seen between CL simulations and RPA structure factors over the full range

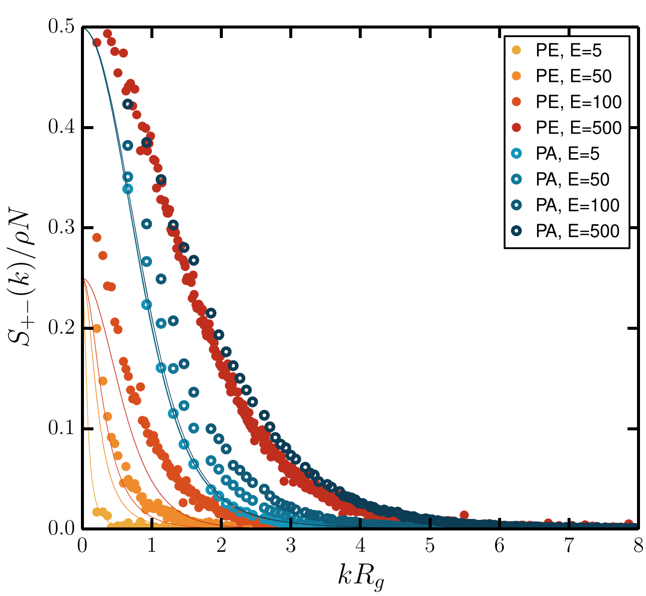

FIG. 9. Dilute-phase opposite-charge structure factors for a symmetric polyelectrolyte (PE) blend (orange-red, CL: filled points, RPA: lines) and an equivalent polyampholyte (PA) diblock (blue, CL: open points, RPA: lines). All structure factors are for $\bar{a} = 0.2$, $B = 1.0$, and $C = 10^{-3}$ and by CL predictions are in the dilute one-phase region of the phase diagrams shown in Fig. 6.

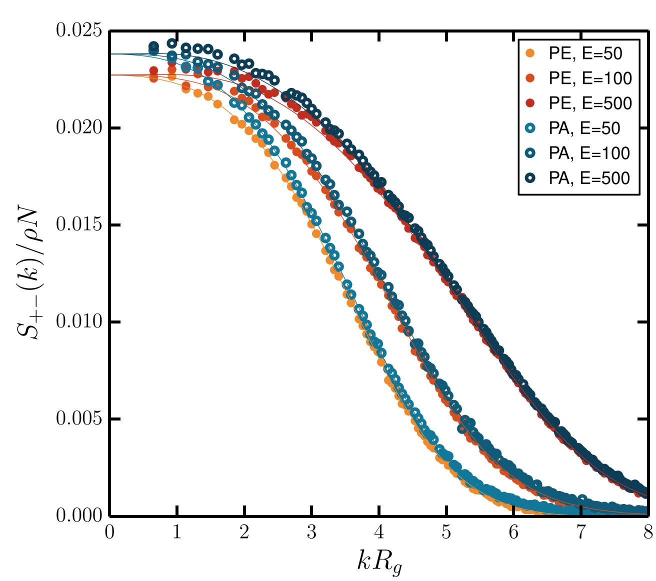

FIG. 10. Concentrated-phase opposite-charge structure factors for a symmetric polyelectrolyte (PE) blend (orange-red, CL: filled points, RPA: lines) and an equivalent polyampholyte (PA) diblock (blue, CL: open points, RPA: lines). All structure factors are for $a = 0.2$, $B = 1.0$, and $C = 10$.

of E, and the PE and PA systems are indistinguishable except for residual E-independent differences as $k \to 0$.

IV. DISCUSSION AND OUTLOOK

By scaling analysis and field-theoretic simulations, we have argued that symmetric polyelectrolyte complex coacervates are thermodynamically analogous to block polyampholyte self-coacervates. In the scaling analysis, the structure of the coacervate phase is insensitive to chain ends and therefore to joining of pairs of polyelectrolyte molecules. According to the random phase approximation, the structure of this phase can be described by a disordered assembly of charges with electrostatic screening length $\xi = (2EC)^{-1/4}$. We have confirmed this prediction using unapproximated FTSs of the concentrated-phase structure factor (Fig. 10), which shows no difference in opposite-charge correlations by joining the polyanions and polycations.

In contrast, the dilute supernatant phase shows important differences under conditions of extreme dilution and/or very low charge density: opposite-charge correlations are largely absent in the polyelectrolyte blend in these limits, but the physical joining of blocks in the diblock polyampholyte ensures that opposite-charge correlations are always present. However, dimers are readily formed for a slight increase in concentration or charge density. Our FTS structure factor data in Fig. 9, combined with observed molecular-weight dependence of the critical point, support the picture that the dilute phase of a polyelectrolyte blend is qualitatively indistinguishable from an equivalent diblock polyampholyte under conditions of charge density, molecular weight, and solvent dielectric constant that are relevant for observing complex coacervation.

The model we have analyzed here is the simplest model that portrays the thermodynamic analogy between polyelectrolyte complex coacervates and block polyampholyte self-coacervates. It is straightforward to extend these simulations to the study of richer phase-separation phenomena, for example, by explicit inclusion of counter ions and added salt[57] or by breaking the symmetry of the charge densities and chain/block lengths considered here. In addition, by appropriate substitution of the single-molecule partition functions, Q, in the field theory model, it is possible to consider other block polyampholyte sequences, other polyelectrolyte-chain architectures, or the inclusion of non-charged blocks or additives.[78] Finally, molecular-species-derived variation in backbone hydrophilicity/hydrophobicity can be modeled with a multi-species matrix[64] of excluded-volume parameters and an explicit polar or polarizable solvent;[79] using such models, a richer variety of thermodynamic behaviors and complex microstructures in the coacervate phase might be accommodated.

SUPPLEMENTARY MATERIAL

See supplementary material for dependence of FTS-CL binodals on the density smearing scale a, and demonstration of finite-size error removal.

ACKNOWLEDGMENTS

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APPENDIX A: GAUSSIAN APPROXIMATION

The model discussed in Sec. III can be analyzed using a Gaussian approximation that is also commonly referred to as the random phase approximation (RPA).[51,57] Here we pursue expressions for $\beta F R_g^3$, $\beta\mu$, and $\beta\Pi R_g^3$ that result from a Gaussian approximation for $\mathcal{Z}_c$. The analysis proceeds, following the methods outlined in Ref. 55, by expanding the single-chain partition functions, Q_i, to second order in field fluctuations around the homogeneous saddle point configuration. The resulting Helmholtz free energy *density* for the symmetric charge-neutral mixture of oppositely charged polyelectrolytes is given by

$$\beta F^{(1)} R_g^3 = C \ln \frac{C}{2} - C + \frac{BC^2}{2} + \frac{1}{4\pi^2}\int_0^\infty dk\, k^2 \ln\left(1 + BCg_D\left(k^2\right) V^2 \hat{\Gamma}^2(k)\right) + \frac{1}{4\pi^2}\int_0^\infty dk k^2 \ln\left(1 + k^{-2} ECg_D\left(k^2\right) V^2\hat{\Gamma}^2(k)\right), \tag{A1}$$

where k is in R_g^{-1} units, and we have used Fourier conventions $f(\mathbf{r}) = \sum_{\mathbf{k}} \hat{f}(\mathbf{k})\, e^{i\mathbf{k}.\mathbf{r}}$, $\hat{f}_k = V^{-1}\int d\mathbf{r}\, f(\mathbf{r})\, e^{-i\mathbf{k}.\mathbf{r}}$. In the RPA free-energy expression, the first three terms describe the mean-field free energy, including ideal-gas entropy, of a homogeneous solution of polymer chains at fixed concentration. The dimensionless polymer chain density (C), excluded volume parameter (B), and electrostatic strength (E) are as defined in Sec. III. $g_D(x) = 2x^{-2}\left[e^{-x} + x - 1\right]$ is the Debye function for an ideal continuous Gaussian chain,[80] while $V^2\hat{\Gamma}^2(x) = \exp\left(-\bar{a}^2 x^2\right)$. As noted previously,[51,55,56] the mean-field free energy for a system with periodic boundary conditions has no E dependence, and any electrostatically driven phase separation is exposed only by considering field fluctuations.

The final term in Eq. (A1) is the RPA fluctuation correction to the electrostatic free energy density, $F_{el}^{(1)}$. While we use its unapproximated form to construct the RPA phase diagram in Fig. 6, the integral can be analysed analytically in two asymptotic concentration limits. By taking the limit $a \to 0$, and making the approximation $g_D(x) \approx 2x^{-1}$, the wave-number integral can be evaluated analytically to yield $\beta F_{el}^{(1)} R_g^3 \approx \sqrt{2}(2EC)^{3/4}/12\pi$, which is accurate for dense systems of overlapping high-molecular-weight polyelectrolyte chains with $R_g \gg a$ and $EC \gg 1$. Restoring dimensional parameters, one finds $\beta F_{el}^{(1)} \sim \left(l_B \rho\sigma^2 b^{-2}\right)^{3/4}$, as stated in the scaling analysis of Sec. II A. In contrast, the extreme dilution limit can be uncovered by considering the expansion $k^2 \ln\left(1 + k^{-2}ECg_D\left(k^2\right)V^2\Gamma^2(k)\right) \approx k^2 \ln\left(1 + k^{-2}EC\right)$

$-EC + ECg_D\left(k^2\right)V^2\Gamma^2(k)$, valid for $EC \to 0$. The first two terms in the expansion describe inter-chain correlations with self-energy excluded, while the last is from electrostatic self-interaction. Evaluating the first two terms of the wave-number integral yields

$$\frac{1}{4\pi^2}\int_0^\infty dk\left[k^2\ln\left(1+k^{-2}EC\right)-EC\right] = -\frac{1}{12\pi}(EC)^{3/2} = -\frac{(4\pi)^{1/2}}{3}\left(l_B N\rho\sigma^2\right)^{3/2}R_g^3, \quad \text{(A2)}$$

as predicted by the Debye-Hückel theory [cf. Eq. (1)].

Chemical potentials and pressures can be obtained from the Helmholtz free energy as

$$\beta\mu^{(1)} = 2\ln\frac{C}{2} + 2BC + \frac{2}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{Bg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)}{1+BCg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)} + \frac{Eg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)}{k^2+ECg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)}\right], \quad \text{(A3)}$$

$$\begin{aligned}\beta\Pi^{(1)}R_g^3 = C + \frac{BC^2}{2} &+ \frac{1}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{BCg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)}{1+BCg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)} - \ln\left(1+BCg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)\right)\right]\\ &+ \frac{1}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{ECg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)}{k^2+ECg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)} - \ln\left(1+k^{-2}ECg_D\left(k^2\right)V^2\hat{\Gamma}^2(k)\right)\right],\end{aligned} \quad \text{(A4)}$$

where the chemical potential is for addition/removal of *neutral pairs* of polyelectrolyte chains.

Similar RPA expressions can be obtained for the diblock polyampholyte as follows:

$$\beta F_{PA}^{(1)}R_g^3 = \frac{C}{2}\ln\frac{C}{2} - \frac{C}{2} + \frac{BC^2}{2} + \frac{1}{4\pi^2}\int_0^\infty dk\,k^2\left[\ln\left(1+BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)\right) + \ln\left(1+k^{-2}ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)\right)\right], \quad \text{(A5)}$$

$$\beta\mu_{PA}^{(1)} = \ln\frac{C}{2} + 2BC + \frac{2}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{Bh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)}{1+BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)} + \frac{Eh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)}{k^2+ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)}\right], \quad \text{(A6)}$$

$$\begin{aligned}\beta\Pi_{PA}^{(1)}R_g^3 = \frac{C}{2} + \frac{BC^2}{2} &+ \frac{1}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)}{1+BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)} - \ln\left(1+BCh_+\left(k^2\right)V^2\hat{\Gamma}^2(k)\right)\right]\\ &+ \frac{1}{4\pi^2}\int_0^\infty dk\,k^2\left[\frac{ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)}{k^2+ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)} - \ln\left(1+k^{-2}ECh_-\left(k^2\right)V^2\hat{\Gamma}^2(k)\right)\right].\end{aligned} \quad \text{(A7)}$$

The joining of neutral pairs reduces the number of molecules by a factor two, leading to modified ideal-gas translational entropy contributions. In addition, as shown below, the pairing produces a different character to electrostatic screening under dilute conditions. These are the dominant differences in comparing RPA predictions of dilute-state thermodynamic properties between the polyelectrolyte blend and diblock polyampholyte. The mean-field interaction terms are unchanged by the pairing, since the overall monomer concentration is unmodified.

The diblock polyampholyte RPA fluctuation corrections contain composite Debye functions $h_\pm(x) = g_D(x) \pm x^{-2}\left(e^{-x}-1\right)^2$, which enforce excluded-volume correlation and electrostatic correlation of the two connected blocks. Note that $\lim_{x\to\infty} h_\pm(x) \sim 2/x$ as for the homopolymer Debye function g_D and that the RPA fluctuation corrections for both PA and PE are quantitatively comparable under concentrated overlapping conditions. Consequently, the high-C asymptotic expression $\beta F_{el}^{(1)} \approx \sqrt{2}(2EC)^{3/4}/12\pi$ produced previously for the symmetric polyelectrolyte blend applies also to the diblock polyampholyte. However, the enforcement of pairing in the extreme dilute limit by the composite Debye functions modifies the $C \to 0$ RPA predictions. In this case, each isolated coil is charge neutral but polarizable, leading to a long-range electrostatic interaction that is of fluctuating-dipole-induced-dipole type rather than the Debye-Hückel form found in the PE system for well-separated charged entities. This behavior is uncovered by noting that $0 < h_-(k^2)/k^2 < 1\ \forall k$, so that for low-$C$ the logarithm in the electrostatic free energy can be expanded in an analytic power series of EC: $\ln\left(1+k^{-2}ECh_-\left(k^2\right)V^2\Gamma^2(k)\right) = -\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\left(k^{-2}ECh_-\left(k^2\right)V^2\Gamma^2(k)\right)^n$. The first-order concentration term is the self-interaction overcounting included in βU_{int}, which should be eliminated by βU_{SI}. To second order in concentration, we find an electrostatic correction to the second virial coefficient: $\beta F_{el}^{(1)} \sim -E^2C^2$, which is $\sim -b\left(l_B\sigma^2\right)^2 N^{5/2}\rho^2$ upon restoring dimensionful quantities, as stated in Sec. II B.

The RPA Eqs. (A2)–(A7) can be used without further approximation to numerically generate RPA phase-coexistence envelopes (solid lines in Fig. 6) by searching for low and high chain densities (C^I and C^{II}) with equal $\beta\mu^{(1)}$ and $\beta\Pi^{(1)}$ so that chemical and mechanical equilibrium of

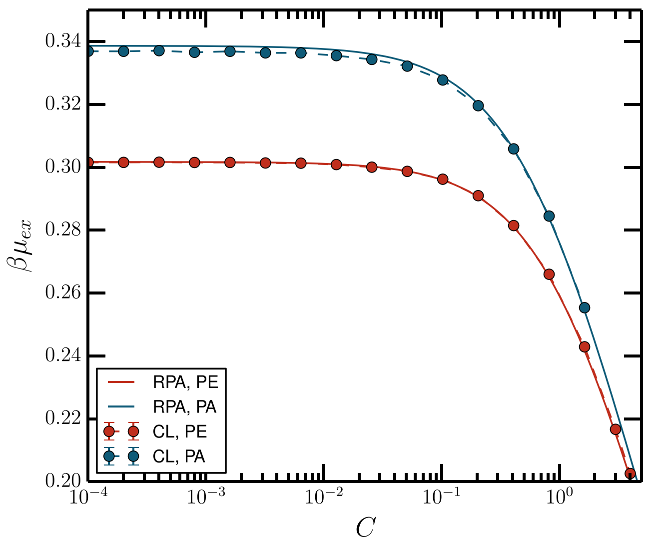

FIG. 11. The excess chemical potential $\beta\mu_{ex} = \beta\mu - \beta\mu_{MFT}$ for $E = 0$. Here electrostatic interactions are suppressed entirely, so that the PE and PA systems differ only by polymer-chain molecular weight. In the $C \to 0$ limit, $\beta\mu_{ex}$ describes the excluded-volume renormalization in a solution of isolated polymer coils without inter-chain interactions.

the coacervate and supernatant phases is achieved. Additionally, we compare the RPA to full FTSs in limits that they are expected to agree. Figures 11–13 show comparisons between FTS-CL and RPA for $\beta\mu$ at both $E = 0$ and $E = 180$ and $\beta\Pi R_g^3$ at $E = 180$. $E = 180$ is lower than, but close to, the critical value for the RPA PE phase diagram (Fig. 6), while it is greater than the PA critical value from RPA analysis. Hence, the signature of a PA compositional instability related to self-coacervation is evident in Figs. 12 and 13. These signatures vanish with FTS-CL sampling due to higher-order fluctuation corrections that suppress phase separation to higher E. In all cases, RPA observables are an excellent match for CL at high C, but the agreement diminishes for very low C, high E, or close to binodal boundaries as expected.

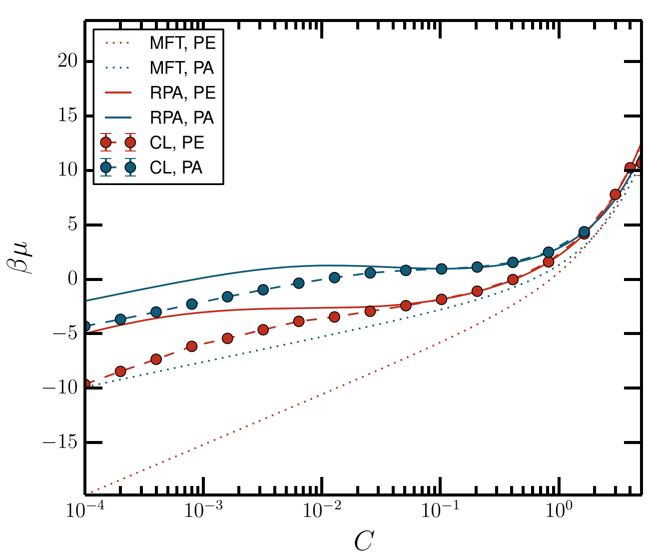

FIG. 12. Chemical potential, including mean-field with ideal-gas contribution, for $E = 180$. Mean-field theory (MFT), CL, and RPA mutually agree for $C \gg 1$. As C is reduced, CL and RPA remain in agreement with $C \lesssim 0.1$ as MFT differs. The PA RPA concentration sweep crosses the phase separated region in Fig. 6; a region of negative curvature is evident that vanishes for CL.

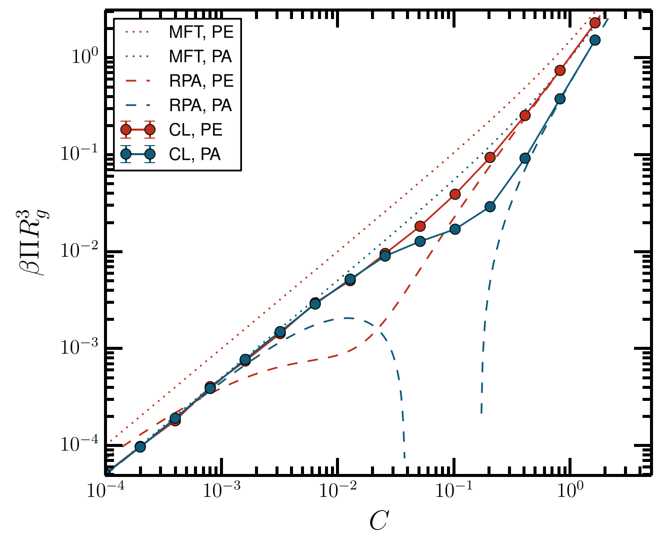

FIG. 13. Osmotic pressure for $E = 180$. The RPA pressure shows a signature of spinodal instability, which is suppressed (to higher E) in CL. Otherwise CL and RPA pressures agree well at high C where the RPA is expected to be good.

APPENDIX B: NUMERICAL DETAILS OF FTS-CL SIMULATIONS

All FTS results reported here were generated using a periodically repeated cubic simulation cell of side L. For denser systems, finite-size errors are easily suppressed and $L = 6.4$ R_g is reliable. Under dilute conditions, larger finite-size errors were evident (see the supplementary material), especially in chemical potentials, due to longer correlation lengths associated with both excluded-volume and electrostatic interactions (recall the RPA weak-fluctuation result of $\xi = (2EC)^{-1/4}$ for the electrostatic correlation length in a homogeneous state) and cells with up to $L = 12.8$ R_g were used. Phase-coexistence curves in Fig. 6 were generated with $L = 9.6$ R_g for PE and $L = 6.4$ R_g for PA. In all the cases, fields were sampled with a spatial collocation mesh of resolution $\Delta x = 0.2$ R_g, which was found to be sufficient to fully resolve chemical potentials and pressures across a wide range of E and C values. The segment smear scale was fixed to $a = 0.2$ R_g and the excluded volume parameter to $B = 1.0$ throughout.

The modified diffusion equations [Eqs. (10) and (11)] were solved using a pseudospectral approach with operator splitting[81–83] and fixed contour step size $\Delta s = 0.01$. The exponential time difference (ETD) algorithm[61,72] with time step $\Delta t = 0.1$ was used to numerically propagate the CL equations of motion [Eqs. (13) and (14)]. All simulations were performed on NVIDIA Tesla M2070 or K80 graphics processing units.[84]

An efficient low-variance pressure operator[61] for the model used here is

$$\tilde{\Pi}\,[\varphi, w] = \sum_l \frac{C\bar{\phi}_l}{\alpha_l} + \sum_l \frac{C\bar{\phi}_l}{\alpha_l}\frac{1}{V} \times \int d\mathbf{r}\left[\frac{2}{3}\Phi_{\nabla}^{(l)}(\mathbf{r}) + \sum_j \Phi_j^{(l)}(\mathbf{r})\left(i\left(\Gamma_2 - \frac{\Gamma}{2}\right)\star w + i\frac{\sigma_j}{\sigma}\left(\Gamma_2 - \frac{\Gamma}{6}\right)\star\varphi\right)\right], \tag{B1}$$

where $\mathbf{r}$ is in R_g units, σ_j is the charge valence of a segment of species j, $\Gamma_2(\mathbf{r}) = \left(1 - r^2/3\bar{a}^2\right)\Gamma(\mathbf{r})$, $\Phi_\nabla^{(l)} = Q_l^{-1}\int_0^{\alpha_l} ds\, q_l^\dagger(\mathbf{r},s)\nabla^2 q_l(\mathbf{r},s)$, and $\Phi_j^{(l)} = Q_l^{-1}\int_{s\in j} ds\, q_l^\dagger(\mathbf{r},s)$ $q_l(\mathbf{r},s)$ is a species-j density field obtained by counting segments of species j from polymer chain type l. Similarly, an efficient operator for the opposite-charge particle-center-coordinate structure factors is

$$\frac{\tilde{S}_{+-}(\mathbf{k};[\varphi,w])}{\rho N} = iV\sum_{l,j}\frac{\bar{\phi}_l}{\alpha_l}\hat{\Phi}_j^{(l)}(\mathbf{k})\left[\frac{w(-\mathbf{k})}{4B} - \frac{k^2\sigma_j\varphi(-\mathbf{k})}{4\sigma E}\right]e^{a^2k^2/2}, \tag{B2}$$

where $\Phi_j^{(l)}(\mathbf{r})$ is the same density-operator function defined in Eq. (B1), $\hat{\Phi}_j^{(l)}(\mathbf{k})$ is its Fourier transform, and $\mathbf{k}$ is the wave vector in units of R_g^{-1}.

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